

Sensorimotor Cortex EEG signal classification using Hidden Markov Models and Wavelet Decomposition

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Abstract—In this paper, we present an offline dynamic classification technique based on wavelet decomposition and Hidden Markov Models (HMMs). We leverage the time-frequency analysis capability of wavelets as well as HMMs ability of modeling dynamic signals to classify the movement of hands based on EEG signals captured from the brain. We report our results with dataset Iib from BCI Competition 2008 provided by Graz University, and show an error reduction of 23% i.e. reduction from 34% to 26% error when compared with the best previously reported technique that used HMMs for brain signal classification. A comparison of performance with various types of wavelets combined with HMMs is also presented in this paper. Since wavelet decomposition increases feature dimensions, we anticipate that further improvement can be achieved by reducing the feature size by exploiting spatial correlation of data from EEG sensors.

Index Terms—Brain Computer Interface (BCI), EEG Classification, Motor Imagery, Hidden Markov Models (HMM), Wavelet Decomposition

I. INTRODUCTION

Loss of limb control caused by accidents or strokes leaves a lot of people suffering in their day to day lives. The most natural way of providing motor ability to such patients is through detection of their neural activity and translating the imagined movement to give control signal to a neuroprosthetic. People suffering from motor disabilities like Amyotrophic Lateral Sclerosis (ALS) [1] are unable to carry out movements but their neural signals can be

used. Motorimagery causes similar activity in brain signals as the actual performance of those movements [2] [3]. Mapping of neural activity can be carried out using a variety of techniques depending on the application. Some of these techniques include Magnetic Resonance Imaging (MRI) [4], CT Scanning, Positron Emission Tomography (PET) Scanning, Near Infrared Spectroscopy (NIRS) [5], Electroencephalography (EEG) etc. These methods leverage on different chemical and electrical processes taking place inside the brain for analysis. However, motor imagery is a time-based process and for its analysis it is critical to have a signal that gives complete temporal information [6]. For the purpose of this study, EEG was selected as a BCI input since it provides high temporal resolution while being non-invasive [7]. This is the reason that EEG is widely used for this application [6].

In this paper, EEG signals produced during motor imagery of right and left hand movements have been used to develop a technique based on Hidden Markov Models (HMM) which is best suited for EEG classification between two kinds of limb movements. The nature of EEG signal is dynamic, having variations in temporal, spatial and spectral domains. Numerous classification methods have been used over the last decade to accurately classify sensorimotor cortex signals. However, these classifiers being static in nature are unable to model the brain signals which are inherently dynamic in nature. The variations are best captured using dynamic classifiers such as HMM. Classification techniques

including Neural Network [15], SVM [16] and distance based classifiers like LDA [17] have been used for EEG signals. However, in [30] HMM outperforms SVM on EEG signals since SVM does not explicitly capture temporal information. In [19], performance of LDA and HMM was compared with HMM performing better than LDA. Reference [18] also shows HMM's superior performance as compared to LDA, SVM and Bayesian classifier. Static classifiers also underperform on high-dimensional signals and are sensitive to over-training in case of noisy signals [6]. In this paper, we further enhance the performance of HMM using appropriate feature extraction techniques.

HMM is a powerful dynamic classification tool being widely used in voice recognition systems and also shows great potential for BCI applications [18] [19]. Despite its potential and efficacy for BCI classification, limited research has been carried out and only a few research papers have been published based on HMM with motor-imagery for left and right hand signals [19] [20]. In [20] 78% classification accuracy was obtained using PCA based features with HMM. In [19] Hjorth based features were used with HMM and average of 85% classification accuracy was achieved. Through literature survey it was observed that along with HMM there is also need for applying pre-processing and feature extraction techniques that best suit the nature and variations of EEG signals.

In this paper, we have further enhanced the performance of HMM classification by using wavelet based features as wavelet decomposition captures temporal and spectral information well. The performance has been compared with Hjorth parameter based features since it has previously provided good results with HMM in left/right hand movement classification regime [6] [19]. In this study, HMM-Wavelet technique provided highest performance with average classification accuracy of 73.6% for three subjects when compared to other techniques. Performance of different wavelet types has also been discussed. Classification error reduction of 23% was achieved when compared with the performance of Hjorth features.

The next section discusses the changes in EEG signals during motorimagery of left and right hand. Section III is Methodology, which documents details of the proposed technique. Dataset details are given under Section IV -

Experimental Setup. And finally, results and conclusion are given in Section V and Section VI respectively.

II. EEG SIGNALS DURING MOTOR IMAGERY

During motor-imagery, EEG signals are captured over sensorimotor cortex [8]. 10-20 system is used as a standard for placement of electrodes over the scalp for picking up EEG signals [9]. The electrodes over which motor imagery signals for right and left hand can be observed are *C3* and *C4* and areas close to *C3* and *C4* [10]. The oscillations in EEG signals can be categorized in particular frequency bands which are associated with different functional and cognitive states of mind. The amplitude of these signals is usually between $0.5\text{-}100\mu\text{V}$ [11]. The frequency bands that are active during motor imagery are μ -band (8-12Hz) and β -band (13-30Hz) [2] [12] [13]. Event Related Synchronization (ERS) and Event Related Desynchronization (ERD) are the rise and attenuation of oscillations in those frequency bands respectively [2]. According to the studies of Professor Pfurtscheller and his team at Graz University, ERD is observed on *C3* during right side movement (contralateral side of the brain), whereas, ERS is observed over *C4* during that time (ipsilateral side of the brain) in μ -band [2] [10]. In-depth analysis of EEG signal variation carried out by this team has become a backbone of EEG classification theory.

III. METHODOLOGY

A. Signal Pre-processing

EEG signals are contaminated by Electrooculogram (EOG) artefacts. These are caused by eye-movements and include eyelid movements as well as retinal activity [21]. The dataset includes signals from three EOG electrodes and information about the trials which were distorted by EOG artefacts. The method followed in this paper for EOG removal is based on the work of Schlogl et al. mentioned in [22]. EOG-free signal i.e $EEG_{[\text{Without EOG}]}$ is calculated as follows:

$$EEG_{[\text{Without EOG}]} = S_{[\text{Contaminated signal}]} - U_{[\text{EOG}]} \cdot b \quad (1)$$

Here, b is the weighting coefficient given by:

$$b = C_{\text{Auto}}^{-1} C_{\text{Cross}} \quad (2)$$

U is the EOG signal from three EOG electrodes and S is the contaminated signal from EEG electrodes. C_{Auto} is the auto-covariance of U and C_{Cross} is the cross-covariance between U and S . In this study S contains data samples from electrodes $C3$, Cz and $C4$ and can be represented as:

$$S = \begin{bmatrix} \tilde{C}3 \\ \tilde{C}z \\ \tilde{C}4 \end{bmatrix} \quad (3)$$

The uncontaminated signal, EEG after implementing EOG removal process can then be represented as follows:

$$EEG = \begin{bmatrix} \tilde{C}3_m \\ \tilde{C}z_m \\ \tilde{C}4_m \end{bmatrix} \quad (4)$$

Here, subscript m with the electrode number identifies the modified signal after EOG removal. EEG was then filtered using band-pass Butterworth filter to extract μ and β -bands collectively to narrow down the spectrum where ERD/ERS potentials are generally more discernible.

B. Feature Extraction

Details on the two feature extraction techniques compared in this paper are given below:

1) *Hjorth Parameters*: Hjorth Parameter is a popular feature extraction technique used with time series data such as speech recognition systems and EEG signals [19] [23]. It is based on three parameters namely Activity, Mobility and Complexity. Activity is simply variance of the amplitude of the signal which represents the mean power. Mobility is the square root of the ratio of variance of first derivative of the time series to its variance and may be perceived as mean frequency. Complexity is the ratio of mobility of the first derivative of the signal to its actual mobility and it represents the deviation of the shape of the signal from sine wave [24].

For each trial, time samples from EEG were divided into non-overlapping windows of equal length, represented by $c_E(n)$, where n is a window number having value 1 to N and E is the electrode that could be $C3$, Cz or $C4$. Hjorth parameters were calculated on each window as shown in the following equations. Activity, Mobility and Complexity have been denoted as Act , Mob and Com respectively.

$$Act_E(n) = var(c_E(n)) \quad (5)$$

$$Mob_E(n) = \sqrt{\frac{Act(c_E(n)')}{Act(c_E(n))}} \quad (6)$$

$$Com_E(n) = \frac{Mob(c_E(n)')}{Mob(c_E(n))} \quad (7)$$

The feature set $\bar{F}_H$ given to the classifier can be represented as:

$$\bar{F}_H = [\bar{H}(1) \quad \dots \quad \bar{H}(n) \quad \dots \quad \bar{H}(N)] \quad (8)$$

where,

$$\bar{H}(n) = \begin{bmatrix} Act_{C3}(n) \\ Mob_{C3}(n) \\ Com_{C3}(n) \\ Act_{Cz}(n) \\ Mob_{Cz}(n) \\ Com_{Cz}(n) \\ Act_{C4}(n) \\ Mob_{C4}(n) \\ Com_{C4}(n) \end{bmatrix} \quad (9)$$

2) *Wavelet Decomposition*: Wavelets are known waveforms that slide over the actual time-based signal and provide correlation coefficients. These wavelet signals can be expanded and contracted i.e. scaled in time. The larger the time scale, the more information on lower frequencies is extracted. The smaller the scale, the more it focuses on higher frequencies and abrupt transitions in the signal [25]. This property makes wavelet decomposition very useful for non-stationary signals since it provides time-frequency analysis. Three types of wavelets similar to the shape of the EEG signals were used – db10, db7 and Sym7. The wavelets are represented by their scaling function ϕ and a wavelet function ψ which provide low and high frequency coefficients respectively. Low-frequency coefficients are termed as Approximations - $A(n)$ while high frequency coefficients are referred to as Details - $D(n)$ [25].

Wavelet decomposition can be carried out upto multiple levels to extract more information from the calculated coefficients. The Wavelet Tree is as shown in Fig. 1. It shows wavelet decomposition up to two levels. At each level, Approximation and Details coefficients are calculated

as given in the following equations.

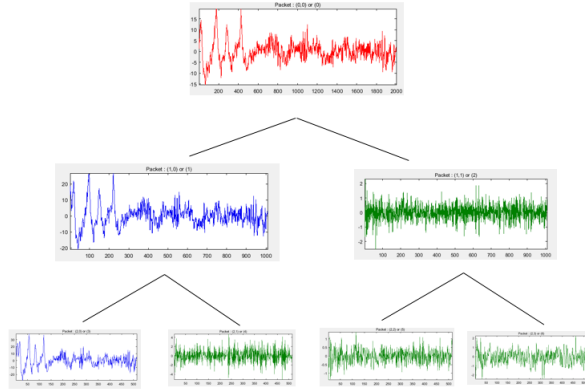


Fig. 1. Wavelet Tree

For Approximation coefficients:

$$\bar{a}_{lE}^i(k) = \sum_{n=1}^N \bar{a}_{(l-1)E}^i(n) \phi_l(n-k) \quad (10)$$

for $k = 0:N-1$

Here, l represents the level of decomposition, E represents the electrode number and n represents time information in the signal where total number of time samples is N , as opposed to Hjorth Parameters where N represented the total number of windows. Shifting of the signal is represented by k . For each level, $2^{l/2}$ number of Approximation vectors are obtained, hence the number of co-efficient vector can be given by i where $i = 1$ to $2^{l/2}$ for each level. Similarly, Detail coefficients can be given by:

$$\bar{d}_{lE}^i(k) = \sum_{n=1}^N \bar{d}_{(l-1)E}^i(n) \psi_l(n-k) \quad (11)$$

for $k = 0:N-1$

Here, $\bar{a}_{0E}^1$ and $\bar{d}_{0E}^1$ is the original signal from EEG and can be given by:

$$\bar{a}_{0E}^1 = \bar{d}_{0E}^1 = [c_E(1) \quad \dots \quad c_E(n) \quad \dots \quad c_E(N)] \quad (12)$$

Here, $c_E(n)$ represents time samples in each electrode obtained from EEG, as opposed to Hjorth Parameters where

$c_E(n)$ represented the windows. In this paper, decomposition was carried out upto three levels and coefficients from the third level were provided to HMM classifier. At the third level, eight features per electrode were obtained. The feature set $\bar{F}_W$, given to classifier can be represented as:

$$\bar{F}_W = \begin{bmatrix} \bar{A}_{3C3}(1) & \dots & \bar{A}_{3C3}(n) & \dots & \bar{A}_{3C3}(N) \\ \bar{D}_{3C3}(1) & \dots & \bar{D}_{3C3}(n) & \dots & \bar{D}_{3C3}(N) \\ \bar{A}_{3C4}(1) & \dots & \bar{A}_{3C4}(n) & \dots & \bar{A}_{3C4}(N) \\ \bar{D}_{3C4}(1) & \dots & \bar{D}_{3C4}(n) & \dots & \bar{D}_{3C4}(N) \end{bmatrix} \quad (13)$$

where,

$$\bar{A}_{3E}(n) = \begin{bmatrix} \bar{a}_{3E}^1(n) \\ \bar{a}_{3E}^2(n) \\ \bar{a}_{3E}^3(n) \\ \bar{a}_{3E}^4(n) \end{bmatrix} \quad (14)$$

$$\bar{D}_{3E}(n) = \begin{bmatrix} \bar{d}_{3E}^1(n) \\ \bar{d}_{3E}^2(n) \\ \bar{d}_{3E}^3(n) \\ \bar{d}_{3E}^4(n) \end{bmatrix} \quad (15)$$

C. Classification

Dynamic classification technique was based on Hidden Markov Model. It was used in an off-line, trial based classification regime. The property of HMM that makes it ideal for time-series data [6] like speech and bio-signals is the fact that it caters for variations in time i.e. it looks for patterns and their occurrence throughout the signal by dividing the signal into 'states'. This technique is based on probabilistic modeling of each class. Detailed explanation of this method has been provided by Rabiner in [27].

Kevin Murphy's HMM tool was used in this study to carry out classification. Two HMMs were designed, one for each class Left and Right - HMM_L and HMM_R, represented by models λ_l and λ_r . Classification was carried out as follows:

For $P(O | \lambda_l) > P(O | \lambda_r)$ - Left Hand movement

For $P(O | \lambda_l) < P(O | \lambda_r)$ - Right Hand movement

Here, $P(O | \lambda)$ is the probability of observation given a model λ . Observation - O is the feature set obtained using the two feature extraction techniques as discussed. O can be given as:

$$O = \{O(1), \dots, O(n), \dots, O(N)\} \quad (16)$$

For Hjorth based features $O(n) = \bar{H}(n)$ and for Wavelet based features:

$$\bar{O}(n) = \begin{bmatrix} \bar{A}_{3C3}(n) \\ \bar{D}_{3C3}(n) \\ \bar{A}_{3C4}(n) \\ \bar{D}_{3C4}(n) \end{bmatrix} \quad (17)$$

The number of states and Gaussian Mixture Models (GMMs) were initially started from a lower value and were increased to improve classification accuracy. After running multiple tests, the number of GMMs and states were set to values that provided highest classification accuracy. Classification accuracy in terms of percentage was calculated.

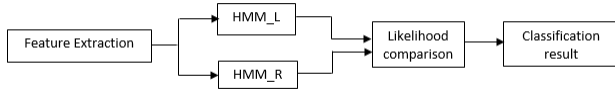


Fig. 2. HMM Classification - Block Diagram

IV. EXPERIMENTAL SETUP

The dataset is based on BCI Competition 2008 - Graz data set II-B [28] and based on a study of Graz University [29]. It comprises of two classes for motor imagery - Left hand and Right hand. First three sessions from the first three subjects of the dataset have been used in this paper. Samples from three electrodes were provided, namely *C3*, *C4* and *Cz*. The readings in the dataset had been recorded with a sampling rate of 250Hz and bandpass filtered between 0.5-100Hz. The starting 8 seconds from each trial were used to extract features. Each session consists of 120 trials out of which 100 were used for training while 20 trials were used for testing.

V. RESULTS

Two feature extraction techniques i.e. Hjorth Parameters and Wavelets were used with HMM and a static classifier - Neural Network (NN) as well. Comparison of all the four approaches is shown in Fig. 3. The results are based on average classification accuracy achieved for three subjects with three recording sessions each i.e. nine sessions.

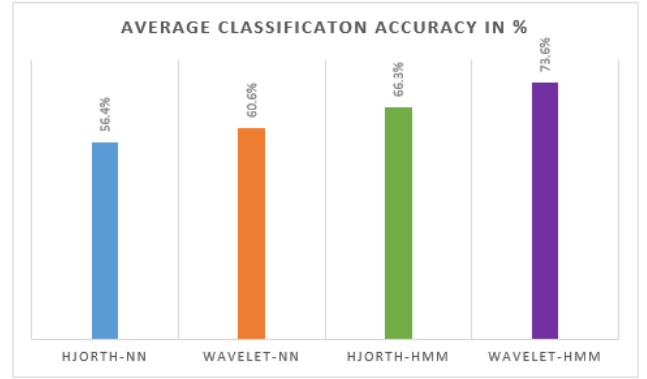


Fig. 3. Average Classification Accuracy

The feature set capturing temporal and spectral information both i.e. wavelet-based features outperformed Hjorth parameters with HMM with 73.6% average classification accuracy. Wavelet-HMM technique provided the highest accuracy of 90% in two sessions for Subject-1. The average accuracy obtained with Hjorth-HMM approach was between 65%-68% for the three subjects. Classification error was reduced by 23% when using wavelet based features with HMM.

The results also depict that HMM performs better than Neural Network - a static classifier on the same set of features. Subject-wise results are given in Fig. 4. Highest accuracy using HMM based techniques was obtained for Subject-1 while the performance for Subject-2 and Subject-3 is similar.

Performance of different wavelet types was also compared with HMM classification. Average classification accuracy of db10, db7 and Sym7 is given in Fig. 5. Db10 provided highest classification accuracy. It captured non-linearity in the signals well.

VI. CONCLUSION

Four classification approaches were applied and compared in this study. Wavelet-HMM approach provided the highest average classification accuracy. This paper shows that the combination of Wavelet Decomposition with HMM classifier holds a lot of potential and is better at catching transitions of signals in time and frequency. Also, the fact that dynamic classifiers are best suited for EEG signals has

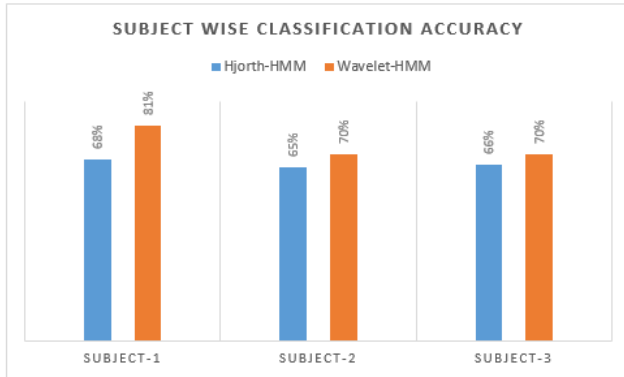


Fig. 4. Subject-wise performance of HMM

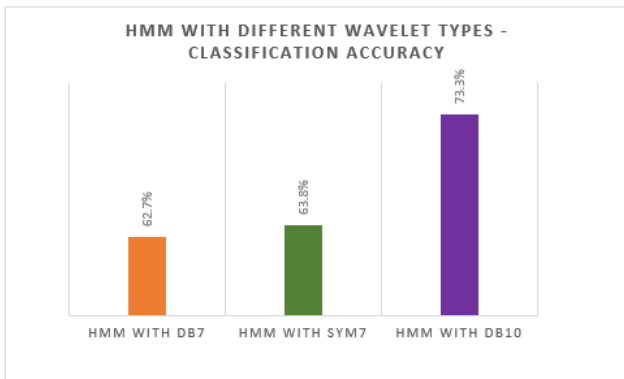


Fig. 5. Comparing Wavelet types with HMM

been proved.

In this paper, results were compared with Hjorth parameter based features with HMM, since Hjorth based features is previously the best known technique with HMM [19] for left/right motor imagery problem. This study shows an average 23% classification error reduction by using Wavelet Decomposition i.e reduction from 34% error (Hjorth-HMM) to 26% error (Wavelet-HMM) on Dataset II-B from BCI Competition 2008 provided by Graz University.

For future work it is assumed that accuracy can be further improved by addition of spatial filters to capture the most discriminating features and also by exploring probabilistic models of the data in-depth.

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